Self-Preference Is Absent in Verifiable Instruction-Following Revision: A Four-Model Test Under Genuine Authorship

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Abstract

Large language models (LLMs) increasingly review and revise text, including their own. A documented *self-preference* bias—models favoring their own generations when acting as judges—raises the question of whether models also resist valid corrections to their own writing. We test this in a setting where "valid" is decided not by another model but by a deterministic verifier: instruction-following revision on IFEval. A model writes a draft; the official IFEval checker confirms the draft violates a constraint and that a candidate edit fixes it; the model then accepts or rejects that edit either as the genuine in-context *author* or as a *fresh* model that sees the draft neutrally. Across four mid-tier model families and 85 author-versus-fresh comparisons, we find no detectable self-preference: authors reject verified-good fixes to their own drafts at essentially the same rate as fresh models judging the same drafts (gap $-5.1\,pp$, 95% CI $[-12.9, +2.7]$). A self-skepticism hint from a smaller pilot did not replicate at scale. The one robust observation is qualitative: when authors do reject a verified-good fix, 97% of their stated reasons are flaw-catching rather than preference—about the *character* of rejections, not an elevated *rate*. Effects smaller than $\sim$13 pp cannot be excluded at this sample size.

1 Introduction

When a language model judges text, it tends to favor text it produced itself. This *self-preference* (or self-bias) is documented in the LLM-as-a-judge literature: models both recognize and prefer their own generations [Panickssery et al., 2024], and the bias is amplified in self-refinement loops [Xu et al., 2024]. A natural and practically important extension is *revision*: when a model is shown a concrete, correct fix to a draft it wrote, is it more reluctant to accept that fix than a model that did not write the draft? If so, agentic "self-review" pipelines would be systematically handicapped on a model's own output.

Answering this cleanly is hard for two reasons. First, "is this fix an improvement?" is usually itself an LLM judgment, which reintroduces the very bias under study (circularity). Second, the comparison must isolate *authorship*: most prior probes use a told label ("you wrote this") rather than the model's genuine generation history.

We address both by restricting attention to *verifiable* instruction following. Using IFEval [Zhou et al., 2023] and its official deterministic checker, "the draft violates constraint c" and "this edit makes the draft satisfy c without breaking any other constraint" are ground-truth facts established by program, not by a model. We then compare two deciders on each verified-good fix: the model that *actually generated the draft in-context* (the *author*) versus a *fresh* model that never saw the generation history. The deterministic checker removes the circularity; the in-context generation removes the told-label confound.

Contribution. We test whether mid-tier LLMs treat verified-good fixes to their *own* instruction-following drafts differently than fixes to another model's draft, under genuine in-context authorship, across four model families, and find *no detectable effect*. The deterministic-checker+author-vs-fresh design is the means that makes this null credible—it is not claimed as a new method, and the result is an *absence in a previously unexamined setting*, not a confirmation of prior work. We report three pilots honestly, including a self-skepticism hint that did *not* replicate, and one careful positive observation about the character (not rate) of author self-rejections.

2 Related Work

It is useful to read prior work as mapping different *cells* of where self-related bias does or does not appear, defined by the task the model performs on the text.

Quality-judging. Self-preference is clearest when a model scores or compares free-form text quality. Panickssery et al. [2024] show models recognize and favor their own generations, with self-recognition correlated with the strength of the bias; Xu et al. [2024] show self-bias is amplified in self-refinement across translation, constrained generation, and reasoning. These establish the phenomenon in the *judging* cell, where the broader LLM-as-judge paradigm [Zheng et al., 2023] is itself subject to a documented *self-enhancement* bias toward a model's own answers.

Answer-selection. When the task is selecting among candidate answers in multi-agent settings, identity-driven effects appear weaker or are dominated by sycophancy, and can be reduced by anonymization [Choi et al., 2026]. This suggests the bias is not uniform across task framings.

Feedback incorporation. On verifiable tasks, models can resist even correct external feedback [Jiang et al., 2025], and sycophancy under user rebuttal pulls in the opposite direction [Kim and Khashabi, 2025]. Crucially, this resistance is documented as a general property, *not* as something specific to a model's own authorship.

Self-correction and session separation. Self-refinement pipelines drive a model to critique and revise its own output [Madaan et al., 2023], but surveys find no consensus that LLMs reliably fix their own mistakes without an external signal [Kamoi et al., 2024], and intrinsic self-correction without ground truth can even degrade reasoning [Huang et al., 2024]. Recent work studies separating production and review sessions to improve error-catching [Song, 2026].

The cell we examine—*verifiable revision under genuine in-context authorship*: does a model accept a machine-verified-correct fix to its own draft less often than a fresh model?—is, to our knowledge, unexamined. That is the gap this paper fills, with a null.

3 Method

Substrate and verifier. We use the real IFEval dataset [Zhou et al., 2023] (541 prompts, each with one or more verifiable instructions such as "respond in all capital letters" or "include the word X"). All pass/fail and fix labels are produced by IFEval's *official* checker (the `instruction_following_eval` instruction registry from google-research), invoked through its strict-evaluation loop. No model is ever used to decide whether a draft satisfies a constraint or whether a fix is correct.

Editable constraints. Because we need fixes that a single minimal edit can make pass, we restrict to a whitelist of *atomically editable* instruction types: keyword inclusion/exclusion and keyword frequency, all-caps / all-lowercase, title in « », postscript, placeholder count, highlighted-section count, end-with-phrase, quotation wrapping, bullet count (≤ 4), and comma avoidance. Count-heavy constraints (minimum sentence/word counts, large letter-frequency targets, paragraph counts) are excluded because they have no clean atomic fix.

Drafts and one verified-good fix. For each prompt, models generate drafts (normal and a "rushed" system-prompt variant, to raise the natural failure rate). We keep drafts that the checker says fail ≥ 1 instruction. For each failing draft we generate targeted minimal edits against the violated constraint's official natural-language description, score every candidate with the checker, and keep *exactly one* candidate that the checker labels **GOOD**: it flips a failed constraint to pass and breaks no passing constraint. Using a single fix per draft eliminates a redundancy artifact observed in an early pilot (Section 4.1). The fix labels GOOD / BAD / NEUTRAL are defined purely by the checker (BAD = breaks a passing constraint; NEUTRAL = no change in outcomes).

Author vs. fresh decision. Each draft+fix is judged under two conditions. *Author (X):* the model that actually produced the draft decides ACCEPT/REJECT *inside its own conversation*—the system prompt, the user task, its own draft as the assistant turn, and then the edit question are all in context. This is genuine authorship, not a told label. *Fresh (Z):* a *different* model, in a new conversation, sees the task, the draft presented neutrally as "a draft response," and the same edit, and answers the same question. The decider is never told the IFEval constraints, that fixes are machine-verified, or which fixes are good; it is asked to judge as an editor and to give a one-line reason. Each condition is run three times at temperature 0.7, with fix order irrelevant (one fix per draft).

Measure. For verified-good fixes define the **self-preference gap**

$$G = r_{\text{author}} - r_{\text{fresh}},$$

where r is the rejection rate of the fix. $G > 0$ means the author rejects valid fixes to its own draft *more* than a fresh model does (treating its own work more harshly—the direction predicted by self-preference/defense, and also by self-skepticism); $G < 0$ means the author is *more* willing to accept corrections to its own work; $G \approx 0$ means no authorship effect. The draft is the unit of analysis (rates are per-draft means over the three reps); we report a 95% CI from a paired bootstrap over drafts (10,000 resamples). We also classify each author rejection's stated reason as *stricter-than-checker* (catching a flaw, e.g. "this still is not fully uppercase") versus *preference* ("I prefer the original"), to characterize *why* authors reject when they do.

4 Results

We ran three pilots of increasing rigor (Table 1). All experiments use OpenRouter-hosted mid-tier models—chosen deliberately, since frontier models nearly ceiling out IFEval and leave few fixable errors. Total API cost was approximately US$1.2.

4.1 P1: told-label proxy (inconclusive, and why)

A first pilot used a told label: a single decider judged the same draft+fixes framed once as "a draft *you* wrote" and once as "a draft written by *another* AI model," byte-identical except one

Pilot	Authorship probe	G (pp)	95% CI (pp)	N
P1: told label, 2 models	"a draft *you* wrote"	-0.7	$\approx \pm 9.5$	35 fixes
P2: genuine, 2 models	in-context author vs. fresh	0.0	$[-13.6, +13.6]$	27 drafts
P3: genuine, 4 models	in-context author vs. fresh	-5.1	$[-12.9, +2.7]$	85 drafts

Table 1: Self-preference gap $G = r_{\text{author/own}} - r_{\text{fresh/other}}$ (rejection-rate difference on verified-good fixes). Positive G would indicate self-preference. P1 is expressed as a rejection gap for comparability (equivalently, acceptance of own-draft fixes was 44.3% vs. 43.5%). All three estimates are at or below zero with CIs spanning zero.

sentence. Pooled over two deciders (gpt-4o-mini, claude-3.5-haiku), acceptance of verified-good fixes was 44.3% (own) vs. 43.5% (other), i.e. $G = -0.7\,pp$ ($\approx \pm 9.5\,pp$), with 35 good fixes across 11 drafts. Two problems made this inconclusive. First, the manipulation is only a *told* label on text the model did not actually author in context. Second, we showed several near-duplicate good fixes per draft, and roughly half of all rejections cited *redundancy* ("already addressed by another edit"), deflating and adding noise to acceptance. We also could not test specificity: zero BAD fixes arose naturally, because atomic edits on independent constraints almost never break a sibling constraint. P1 motivated the design changes in P2/P3 (genuine authorship; one fix per draft) and is reported only as the development path.

4.2 P2: genuine authorship, two models (clean null)

P2 fixes both problems: the author decides inside its own generation conversation, and exactly one verified-good fix is shown per draft. Across 27 drafts (15 gpt-authored, 12 haiku-authored), authors and fresh models accepted good fixes at an *identical* 84.0%, so $G = 0.0\,pp$, 95% CI $[-13.6, +13.6]$. The per-draft signs are symmetric (4 in the self-preference direction, 4 opposite, 19 tied)—no directional lean under the mean. The jump from P1's $\sim$44% to 84% acceptance confirms that the low P1 baseline was the redundancy artifact, not authorship. The only residue was a weak asymmetry in the paired fingerprint: 6 drafts where the author rejected its own good fix while a fresh model accepted it, versus 4 in the mirror direction—a hint of mild *self-skepticism* (authors a touch harsher on their own work), which P3 was designed to power up and test for generality.

4.3 P3: genuine authorship, four models (the powered test)

P3 scales to 85 usable drafts across four mid-tier families: `gpt-4o-mini`, `claude-3.5-haiku`, `gemini-2.5-flash-lite` (substituted for an unavailable `gemini-flash-1.5`), and `llama-3.3-70b-instruct`. Each author is a draft's true generator; the fresh decider is a different family, rotated so each family serves as both author and comparator.

Pooled, authors rejected verified-good fixes to their own drafts 15.3% of the time versus 20.4% for fresh models on the same drafts: $G = -5.1\,pp$, 95% CI $[-12.9, +2.7]$, which includes zero. The self-skepticism hint from P2 did *not* replicate; if anything the point estimate is slightly negative (authors marginally *more* willing to accept fixes to their own work). The paired counts are balanced-to-mirror-leaning: 9 author-rejects-fresh-accepts versus 10 in the mirror direction (any-rep).

Cross-model (lack of) generality. Table 2 shows two families with small positive gaps (gpt, gemini) and two with negative gaps (haiku, llama); no sign holds in a majority of families. The dispersion is almost entirely on the *fresh* side: author self-rejection sits near 15% for all four families,

Author family	N	author rej.	fresh rej.	G (pp)	95% CI (pp)
gpt-4o-mini	34	13.7%	8.8%	+4.9	[−4.9, +14.7]
gemini-2.5-flash-lite	19	17.5%	15.8%	+1.8	[−17.5, +19.3]
claude-3.5-haiku	19	15.8%	29.8%	−14.0	[−29.8, 0.0]
llama-3.3-70b-instruct	13	15.4%	43.6%	−28.2	[−53.8, −7.7]
Pooled	85	15.3%	20.4%	−5.1	[−12.9, +2.7]

Table 2: P3 per-author-family self-preference gap. Two families are positive, two negative; there is no cross-family majority for either sign. Note that *author* rejection rates are tightly clustered (13.7–17.5%) while *fresh* rejection rates vary widely (8.8–43.6%): the per-family gaps are driven mainly by how strict each author's *comparator* families happened to be, not by the authors themselves (Section 5, limitation (c)).

Author-rejection reason class	Count	Share
Stricter-than-checker (flaw-catching)	38	97.4%
Preference ("prefer the original")	1	2.6%
Other	0	0.0%
Total author self-rejections	39	—

Table 3: Reason classification for author self-rejections in P3 (classifier: gpt-4o-mini, fixed rubric; all raw reasons released).

whereas the comparator rejection rate ranges from 8.8% to 43.6%. The apparent per-family "effects" are therefore largely comparator-strictness artifacts of the design (Section 5).

What survives: the character of self-rejections. Of the 39 author self-rejections of verified-good fixes, 38 (97.4%) were classified as *stricter-than-checker*—the author objects that the fix is inadequate or breaks something the verifier/fresh model let through (e.g. "the edit includes the name but breaks the limerick's AABBA scheme")—and only 1 was mere preference (Table 3). So *when* an author rejects a fix to its own work, it is almost never "I prefer mine"; it is flaw-catching. This is a statement about the *character* of rejections, not about their *rate*, which (Table 2) is not elevated relative to fresh models.

5 Discussion and Limitations

Reading across the three pilots. A told-label proxy (P1), a clean two-model authorship test (P2), and a powered four-model test (P3) all return self-preference gaps at or below zero with CIs spanning zero. In verifiable instruction-following revision, mid-tier models neither defend nor disproportionately distrust their own writing: they accept machine-verified corrections to their own drafts at roughly the same high rate (∼80–85%) as a fresh model. Relating this to Section 2: self-preference is documented in quality-*judging* [Panickssery et al., 2024, Xu et al., 2024], appears weak/rare in answer-*selection* [Choi et al., 2026], and resistance to correct feedback exists on verifiable tasks but is not authorship-specific [Jiang et al., 2025]. Our cell—verifiable *revision* under genuine authorship—shows no detectable authorship effect. The bias is cell-dependent, and this cell appears clean.

Limitations. *(a) Mid-tier models only.* We deliberately used mid-tier models to avoid IFEval's ceiling effect; frontier models are untested and could differ. *(b) Low-latitude constraints only.* IFEval constraints are atomic and machine-verifiable; holistic or high-latitude revision (style, argumentation, factuality) is exactly where judging-cell self-preference is strongest and is untested here. *(c) Comparator confound.* In P2/P3 the fresh decider is a *different* family with its own baseline strictness, so per-family gaps conflate authorship with comparator strictness. This is visible in Table 2: author rejection is flat near 15% across families while fresh rejection ranges 8.8–43.6%, so the large negative gaps for claude and llama reflect strict comparators, not lenient authors. A clean deconfounded design would have *all* non-author families judge each draft as fresh. *(d) Power floor.* With $N = 85$ the pooled CI is $\sim \pm 8$–$13\,pp$; effects smaller than ∼13 pp cannot be excluded. We therefore claim "weak or absent," never "proven absent."

The flaw-catching observation, stated carefully. The one robust positive finding is that author self-rejections are overwhelmingly flaw-catching (97.4% stricter-than-checker), not preference-driven. This does *not* say authors are net more self-critical—their rejection *rate* is not higher than fresh models'. It says that the rare self-rejections that do occur are reasoned objections to the fix, not ego. We flag this as a hypothesis for future, properly powered and deconfounded work, not as a demonstrated reversal; the self-skepticism rate signal from P2 did not replicate in P3.

6 Conclusion

In verifiable, mid-tier instruction-following revision under genuine in-context authorship, we find *no detectable self-preference*: models accept machine-verified corrections to their own drafts about as readily as a fresh model does (P3 gap $-5.1\,pp$, 95% CI $[-12.9, +2.7]$), with no cross-family majority for either direction and effects below ∼13 pp unexcludable. A self-skepticism hint from a smaller pilot did not replicate at scale. Placed against prior work, this maps a clean cell: self-preference documented in quality-judging and weak in answer-selection is, in this verifiable-revision setting, not detected. The lone durable observation is qualitative—when authors do reject a verified-good fix, 97% of reasons are flaw-catching rather than preference. Future work should test frontier models, holistic (high-latitude) revision, and a fully deconfounded all-families-as-fresh comparator before drawing rate-level conclusions.

Reproducibility

All code, the official IFEval checker wrapper, fixed random seeds, and every raw decision (author and fresh responses, one-line reasons, and checker labels) for all three pilots are released. The deterministic checker means the verified-good/bad labels are reproducible exactly; model decisions are stochastic (temperature 0.7, 3 reps). Total API cost was ≈US$1.2. See the repository `README` for run order and the environment note (the dataset is loaded directly from the HuggingFace JSONL to avoid a `datasets`/`fsspec` incompatibility).

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